

Trade Bot Advisor v4.0

Multi-Agent + Dynamic Optimizer + Advanced Reasoning

Product Documentation

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<https://huggingface.co/spaces/SutskeverFanBoy/trade-bot-advisor>

DISCLAIMER: Educational and research purposes only. Not investment advice.

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1. Executive Summary

Trade Bot Advisor v4.0 is a multi-agent algorithmic trading signal system with self-optimizing strategy weights and advanced contextual reasoning. It represents a major evolution from single-engine signal generation to a collaborative agent architecture where specialized agents analyze different aspects of the market independently, and an orchestrator synthesizes their findings into a final decision.

Key v4.0 additions over v3.5:

- Multi-Agent System: 4 specialized agents (Analyst, Sentiment, Risk, Orchestrator) with configurable weights and risk veto capability
- Dynamic Strategy Optimizer: Uses ground truth data to learn which indicators perform best, then auto-adjusts voting weights
- Advanced Reasoning: 5 contextual analyses (market regime, liquidity traps, divergences, macro context, exhaustion signals) that adjust confidence beyond what raw indicators show

The system now has 12 interactive tabs, 17 files totaling ~240KB, and processes data from yfinance, Binance, and a built-in sample generator that guarantees the demo always works.

2. System Architecture

6-layer architecture with clear separation:

Layer 1 - Data (data_ingest.py): yfinance + Binance REST + sample fallback. Never returns None.

Layer 2 - Signal Engine (signal_engine.py): 5-indicator voting (RSI, SMA, MACD, Bollinger, Volume) + volatility filter + confidence scoring + ATR-based position sizing.

Layer 3 - Multi-Agent (agents.py): 4 independent agents (Analyst, Sentiment, Risk) + Orchestrator that merges with weighted scoring. Risk Agent has veto power.

Layer 4 - Optimization (optimizer.py + evaluator.py): Ground truth evaluation per indicator. Measures historical accuracy. Computes adaptive weights.

Layer 5 - Reasoning (reasoning.py): Market regime detection, liquidity trap detection, RSI divergence, macro context, exhaustion signals. Adjusts confidence.

Layer 6 - Presentation (app.py, charts.py): 12-tab Gradio UI, Plotly charts, JSON output.

File Structure (17 files, ~240KB)

File	Size	Purpose
app.py	51KB	Main Gradio app, 12 tabs, all UI functions
core/signal_engine.py	23KB	Deterministic 5-indicator signal generator
core/agents.py	17KB	NEW: Multi-agent system (4 agents + orchestrator)
core/optimizer.py	8KB	NEW: Dynamic strategy weight optimizer
core/reasoning.py	16KB	NEW: Advanced contextual reasoning engine
core/backtester.py	9KB	Walk-forward backtesting
core/evaluator.py	8KB	Ground truth evaluator
core/portfolio.py	11KB	Stateful portfolio simulator
core/data_ingest.py	6KB	Data fetch: yfinance + Binance + fallback
core/charts.py	12KB	Plotly chart builders (8 chart types)
core/llm_interpreter.py	5KB	Optional LLM explanation layer

3. Core Modules

3.1 Signal Engine (signal_engine.py)

The deterministic brain. 5-indicator voting: RSI(14), SMA(20/50) crossover, MACD(12,26,9), Bollinger Bands(20,2), Volume ratio. Voting: 3+/5 bullish=BUY, 4+=STRONG_BUY, same for bearish. Extreme volatility filters out BUY signals. Confidence: 0-100 based on agreement, RSI confirmation, volume, volatility. Position sizing: ATR-based SL/TP with risk % per tolerance level.

3.2 Multi-Agent System (agents.py) — NEW v4.0

4 specialized agents, each producing independent AgentReport with signal (BULLISH/BEARISH/NEUTRAL), confidence (0-100), reasoning chain, and data payload.

AnalystAgent

Runs the full signal engine. Extracts technical indicators, votes, bullish/bearish counts. Pure quantitative analysis.

SentimentAgent

Analyzes market mood via: 5-day/20-day price momentum, volume trend (rising/falling/average), volatility expansion/contraction. Produces composite sentiment score. In production: would integrate FinBERT on real news headlines.

RiskAgent

Evaluates: annualized volatility, recent drawdown from peak, ATR gap risk, trend strength (ADX proxy). Produces verdict: APPROVE (safe), CAUTION (reduce size), REJECT (don't trade). Risk score 0-5+.

OrchestratorAgent

Merges all 3 reports using configurable weights (default: Analyst 50%, Sentiment 25%, Risk 25%). Scores: BULLISH=+1, NEUTRAL=0, BEARISH=-1, weighted by confidence. Risk veto: REJECT overrides BUY to HOLD. CAUTION halves conviction. Consensus detection: UNANIMOUS/MAJORITY/SPLIT affects confidence bonus.

3.3 Dynamic Strategy Optimizer (optimizer.py) — NEW v4.0

Self-improving loop: walks through 2 years of historical data, day by day. At each day, measures each indicator's individual accuracy against N-day future returns. Computes per-indicator accuracy (RSI: X%, SMA: Y%, MACD: Z%, Bollinger: W%, Volume: V%). Generates optimized weights proportional to accuracy (minimum 5% weight per indicator). Supports multi-ticker averaging.

Example output: Default weights RSI=20%, SMA=20%, MACD=20%, Bollinger=20%, Volume=20%. Optimized: RSI=15%, SMA=15%, MACD=8%, Bollinger=13%, Volume=49% — because Volume had 70% accuracy while MACD had only 44%.

3.4 Advanced Reasoning (reasoning.py) — NEW v4.0

5 contextual analysis modules that go beyond 'RSI is oversold':

- Market Regime Detection: Classifies as STRONG_UPTREND, STRONG_DOWNTREND, VOLATILE, RANGING, or TRANSITIONAL. Based on up-day percentage, volatility, 20-day price range.
 - Liquidity Trap Detection: Bull trap = price near highs + declining volume. Bear trap = price near lows + declining volume. Warns when breakout may be fake.
 - RSI-Price Divergence: Bearish divergence = price near highs + RSI overbought. Bullish divergence = price near lows + RSI oversold. Signals weakening momentum.
 - Macro Context: Asset-class-aware. Crypto: risk-on/risk-off. BIST: TRY pressure/TCMB. US stocks: goldilocks/correction/uncertain. Commodities: inflation-sensitive.
 - Exhaustion Detection: Extreme RSI + volume spike = buying/selling exhaustion. Potential top/bottom forming.
- Synthesis engine adjusts confidence: regime alignment (+/-10-15%), trap warnings (-10%), divergence (-5%), exhaustion (-8%). Final verdict: STRENGTHENED, CONFIRMED, or WEAKENED.

3.5 Backtesting Engine (backtester.py)

Walk-forward backtest with same signal logic. Configurable RSI thresholds, SMA/vol filters, holding period, evaluation period. Metrics: accuracy, total return, Sharpe, max drawdown, profit factor. Equity curve tracking. Buy & hold comparison.

3.6 Ground Truth Evaluator (evaluator.py)

For every historical signal: records actual price change at 5, 10, 20 day horizons. Per-signal verdict: correct/incorrect. Aggregate: accuracy per horizon, BUY vs SELL accuracy, directional return. Calibration: high-conviction (extreme RSI) vs low-conviction accuracy comparison.

3.7 Portfolio Simulator (portfolio.py)

In-memory stateful portfolio. Opens/closes positions based on signals. Tracks: cash, equity, PnL, win rate, profit factor, max drawdown. Risk controls: max 95% utilization, no duplicates, SL/TP tracking. Supports multi-ticker simulation.

3.8-3.10 Supporting Modules

Data Ingestion (data_ingest.py): 3-tier fetch (Binance for crypto, yfinance for stocks, sample fallback). Never crashes.

Chart Builder (charts.py): 8 Plotly chart types — candlestick+SMA+volume, RSI panel, MACD panel, equity curve, signal map, portfolio equity, PnL distribution, drawdown.

LLM Interpreter (llm_interpreter.py): Optional HF Inference API (Qwen 72B). Template fallback when no token.

Explains signals, never generates them.

4. User Interface — 12 Tabs

1. Analiz

Single-ticker: candlestick, RSI/MACD, structured JSON, interpretation

2. Backtest

Configurable strategy backtesting with equity curve, signal map, metrics

3. Karsilastirma

2-5 ticker side-by-side comparison table

4. Explainability

Why this trade? Indicator breakdown, vote count, trade details

5. AI Decision

Multi-ticker pipeline: signal + backtest validation + portfolio execution

6. Evaluation

Dashboard: last 50 signals, equity curve, drawdown, signal distribution

7. Portfoy Sim

Stateful portfolio: open/close trades, PnL tracking, equity curve

8. Ground Truth

Model ne dedi? Gercek ne oldu? Per-signal 5/10/20 day outcomes

9. Multi-Agent (v4.0)

4-agent analysis with adjustable weights and reasoning chain

10. Optimizer (v4.0)

Dynamic indicator weight optimization from ground truth

11. Reasoning (v4.0)

Contextual analysis: regime, traps, divergence, macro, exhaustion

12. Rehber

Usage guide, signal rules, supported assets

5. Technical Indicators

RSI(14)

Wilder smoothing. >70 overbought, <30 oversold, >60 bullish, <40 bearish.

SMA(20/50) Crossover

Golden cross: SMA20>SMA50 + price above = STRONG_BULLISH. Death cross: opposite.

MACD(12,26,9)

EMA12-EMA26=MACD line. 9-period signal. Histogram crossovers detected.

Bollinger(20,2)

20-SMA +/-2 std. Price above upper=overbought, below lower=oversold.

Volume Ratio

Current/20-day avg. >1.5x=bullish confirm, <0.5x=weak signal.

ATR(14)

Average True Range for SL/TP distance and position sizing.

Volatility Regime

Annualized from 30-day returns. LOW<15%, MOD<30%, HIGH<50%, EXTREME>50%.

6. Signal Rules

5-indicator majority vote: 3+/5 bullish=BUY, 4+=STRONG_BUY. Same for bearish=SELL. Extreme vol + BUY -> HOLD (safety). Confidence: base 50, +25 for 4+ agreement, +/-10 RSI confirm, +5 volume, -15 extreme vol.

7. Risk Management

Conservative: 1% risk, 1.5x ATR SL, 2x TP. Moderate: 2%, 2x, 3x. Aggressive: 4%, 3x, 4.5x. Max position: 20% portfolio. In multi-agent: Risk Agent can VETO (override BUY to HOLD) or CAUTION (halve conviction).

8. Data Sources

Primary: yfinance (stocks/ETFs/commodities/forex). Secondary: Binance REST (crypto, no API key). Fallback: seeded random walk sample data — guarantees demo always works.

9. Supported Assets

Type	Format	Examples
US Stocks	TICKER	AAPL, NVDA, MSFT, GOOGL, TSLA, AMZN, META
BIST	TICKER.IS	THYAO.IS, GARAN.IS, AKBNK.IS
Crypto	TICKER-USD	BTC-USD, ETH-USD, SOL-USD
Commodities	Futures	GC=F (Gold), SI=F (Silver)
ETFs	TICKER	SPY, QQQ
Forex	XXX=X	USDTRY=X

10. v4.0 New Features Deep Dive

A. Multi-Agent Architecture

Problem: Single signal engine can't distinguish between 'technical says BUY' and 'market conditions are too risky for BUY'. Solution: Split into 3 specialized agents + orchestrator.

Analyst: Pure technical analysis via signal engine. Sentiment: Price momentum + volume trends + volatility momentum. Risk: Volatility regime + drawdown + ATR risk + trend strength.

Orchestrator merge: $\text{weighted_score} = \sum(\text{signal_value} * \text{weight} * \text{confidence})$ per agent. Risk veto: if Risk says REJECT, BUY -> HOLD. CAUTION halves conviction.

UI: Adjustable weight sliders for each agent. Full reasoning chain visible.

B. Dynamic Strategy Optimizer

Problem: Fixed equal weights (20% each) ignore that some indicators work better in current conditions. Solution: Walk through 2 years of history. At each day, test each indicator independently. Compute accuracy per indicator.

RSI accuracy: Out of N days RSI said bullish, how many times did price go up in 10 days? Same for SMA, MACD, Bollinger, Volume.

Weight formula: $\text{weight} = \max(0.05, (\text{accuracy} - 40) / 100)$, then normalize to sum=1. Minimum 5% weight prevents any indicator from being fully ignored.

Multi-ticker: Run on multiple tickers, average the weights across all tickers for a more robust estimate.

C. Advanced Reasoning Engine

Problem: 'RSI is 32, signal is BUY' doesn't answer 'Is this a liquidity trap?' or 'Does the macro environment support this?'. Solution: 5 contextual analysis modules.

1. Market Regime: Is it trending, ranging, or volatile? Ranging markets -> mean-reversion signals better. Trending -> trend-following better.
2. Liquidity Trap: Price near highs + volume declining = potential bull trap. Warns before fake breakouts.
3. RSI Divergence: Price making new highs but RSI overbought = bearish divergence. Weakening momentum.
4. Macro Context: Asset-class-specific. Crypto in risk-off? BIST under TL pressure? US in goldilocks or correction?
5. Exhaustion: Extreme RSI + volume spike = climactic buying/selling. Potential reversal.

Synthesis: Adjusts original confidence. Regime alignment: +/-10-15%. Warnings: -5 to -10% each. Final verdict: STRENGTHENED/CONFIRMED/WEAKENED.

11. Performance Metrics

Backtesting: accuracy, total return, Sharpe ratio, max drawdown, profit factor, avg win/loss.

Portfolio: equity, PnL, win rate, profit factor, max drawdown.

Ground Truth: per-horizon accuracy (5d/10d/20d), BUY vs SELL, directional return, calibration.

Optimizer: per-indicator accuracy, optimized vs default weights, ranking.

Multi-Agent: per-agent signal/confidence/weight/contribution, consensus type, reasoning chain.

12. Limitations

- No real-time streaming (point-in-time analysis)
- In-memory state (resets on Space restart)
- Sample data fallback (synthetic when live unavailable)
- No short selling in portfolio sim
- Daily timeframe only (no intraday)
- No fundamental data (P/E, earnings)
- No real news/sentiment API (simulated via momentum)
- Sentiment agent uses price-based proxy, not FinBERT on headlines
- Optimizer recomputes each time (no persistent learned weights)
- Risk Agent thresholds are fixed (not adaptive)

13. Future Roadmap

High Priority

- Persistent SQLite for portfolio + learned weights
- Real FinBERT sentiment on DuckDuckGo headlines
- Multi-timeframe analysis (daily + 4h + 1h)
- Walk-forward parameter optimization

Medium

- Real macro data integration (TCMB, Fed, yields)
- Options flow as signal modifier
- Telegram/Discord alert notifications
- REST API for programmatic access

Exploratory

- RL for dynamic position sizing
- Ensemble strategies
- Cross-asset correlation signals
- Mobile app

14. Tech Stack

Python 3.13 | Gradio 6.x (ssr_mode=False) | Plotly 6.x | yfinance 1.3+ | Binance REST | NumPy | Pandas | Pydantic 2.x | HF Inference API (optional, Qwen 72B) | HF Spaces (free tier)

Total: 17 files, ~240KB. Zero config: git clone + pip install + python app.py